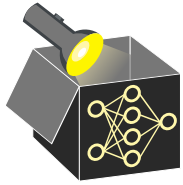
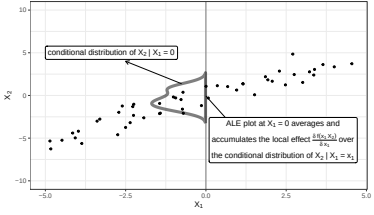
Interpretable Machine Learning



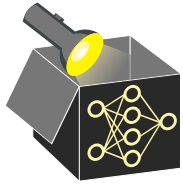
Feature Effects
ALE: Limitations and Follow-up



Learning goals

- Recognize three limitations of standard ALE plots
- Understand the ideas behind DALE and RHALE

BEYOND ALE: THREE ISSUES TO KEEP IN MIND

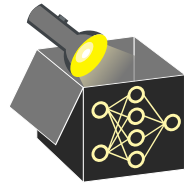


Recall: ALE averages local prediction changes, then accumulates and centers them. This helps handle dependent features. ▸ "Apley, Zhu" 2020

Issue	Why it matters	Follow-up idea
Boundary evaluations	Artificial inputs can still be unrealistic; repeated evaluations cost time.	**DALE:** Gradients at data
Predefined bins	The binning can hide changes or produce noisy estimates.	**RHALE:** Adaptive bins
One average curve	Opposing local effects can cancel each other out.	**RHALE:** Effect heterogeneity

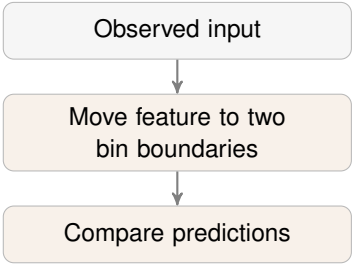
The follow-up work improves how local effects are **estimated and summarized**, while retaining ALE's core idea.

DALE: TAKE LOCAL SLOPES AT OBSERVED POINTS

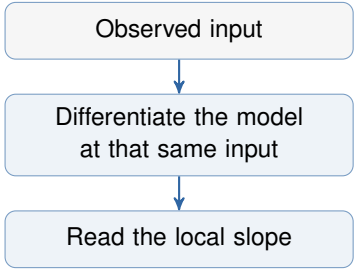


Issue: Standard ALE moves observations to bin boundaries. With dependent features, wider bins can move these inputs further from observed data.

Standard ALE estimation



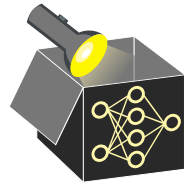
DALE estimation



Why this helps: No artificial evaluation points; automatic differentiation can obtain slopes for many features efficiently. Compute the slopes once, then reuse them for different binnings before averaging and accumulating. ▸ "Gkolemis et al. (DALE)" 2023

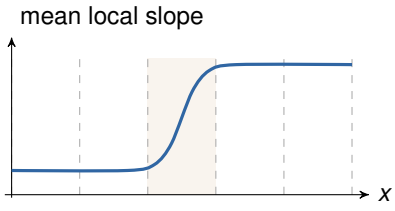
Requirement: A differentiable model with accessible gradients. Wide bins can still bias the approximation.

RHALE: LET LOCAL EFFECTS GUIDE THE BINS



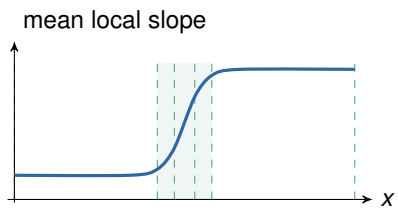
Issue: Wide bins can blur changes; narrow bins may contain too few observations for a stable estimate.

Predefined bins



One bin spans a sharp change

Adaptive bins

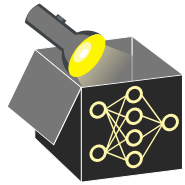


Finer bins around the change

Schematic: same mean local slopes (blue), different bin boundaries (dashed).

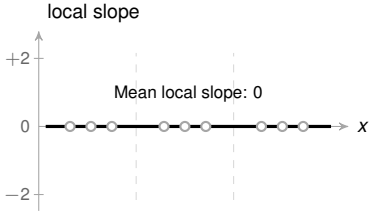
- **How:** RHALE uses DALE's slopes to choose wider bins where the mean slope barely changes and finer bins where it changes quickly, requiring enough observations per bin.
- **Why:** Mixing different mean slopes in one bin can inflate the estimated heterogeneity. Adaptive bins help separate this binning error from genuine differences between local effects. ▸ "Gkolemis et al. (RHALE)" 2023

SAME ALE CURVE, VERY DIFFERENT MODEL BEHAVIOR



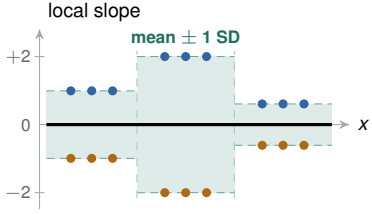
Both models yield a flat ALE curve (mean local slope is 0 everywhere). Does x have no effect, or do opposing effects cancel out?

Case 1: No Local Effect



Zero spread
No local effect on predictions

Case 2: Opposing Local Effects



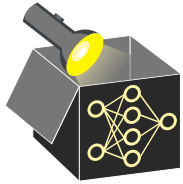
Mean 0, spread $>$ 0
Slopes of both signs cancel

Toy example: on the right, slopes $+a$ and $-a$ are equally common in each bin, so mean $= 0$ and SD $= a$.

Takeaway: A zero average can hide strong effects. **RHALE** reports local-slope spread next to the mean to separate cancellation from no effect.

▸ "Gkolemis et al. (RHALE)" 2023

TAKE-AWAY



Three ideas to remember; three entry points for further reading:

- **DALE: Differential Accumulated Local Effects.**
  Use gradients at observed inputs. ▸ "Gkolemis et al. (DALE)" 2023
- **RHALE: Robust and Heterogeneity-aware ALE.**
  Adapt the bins and report effect heterogeneity. ▸ "Gkolemis et al. (RHALE)" 2023
- **Regional methods: explain effects within subgroups.**
  Look for regions with more homogeneous effects. ▸ "Gkolemis et al. (Effector)" 2024

In practice: Gradient-based estimation needs a differentiable model with accessible gradients. Standard ALE remains useful for prediction-only black boxes, including tree ensembles.

Interpretation: These methods describe the fitted model over the data distribution; causal claims need separate justification.

Try it: The `effector` package provides implementations and examples. ▸ "effector" n.d.